

Introduction to Computer Vision

Aniruddha

What is computer vision?

Computer vision is a sub-field of Artificial Intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs in a way that mimics human vision.

Applications of computer vision

- Autonomous vehicles: Detect and recognize objects such as pedestrians, other vehicles, traffic signs, and lane markings on the road.
- Medical imaging: Analyze medical images such as X-rays, MRIs and CT scans to assist in diagnosis through detection of fractures or tumors with high precision.
- Facial recognition: Used in security systems, smartphones and digital applications for identifying and verifying individuals based on their facial features.
- Retail and e-commerce: Used for product recognition, inventory management, and “just walk out” technology in stores.
- Agriculture: Used for crop monitoring, pest detection, and yield estimation in agriculture.



Figure 1: Credits: Dllu, wikimedia commons

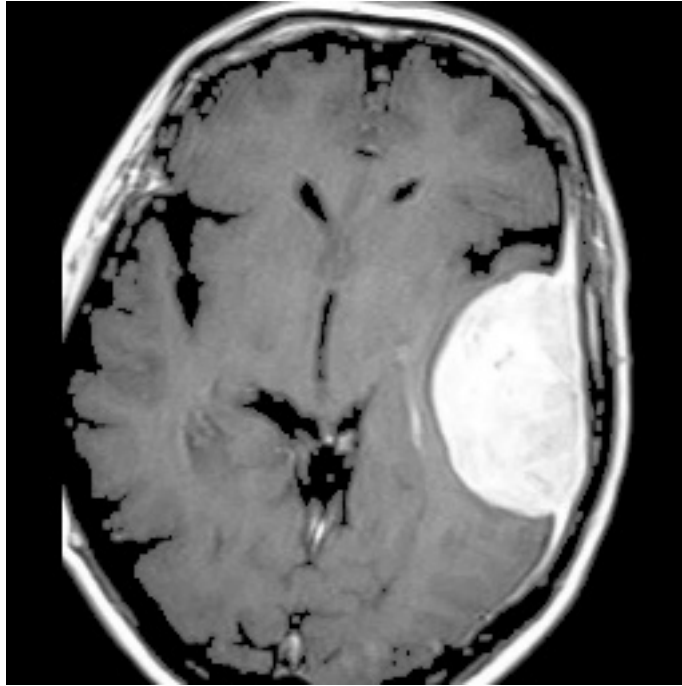


Figure 2: Credits: Roboflow

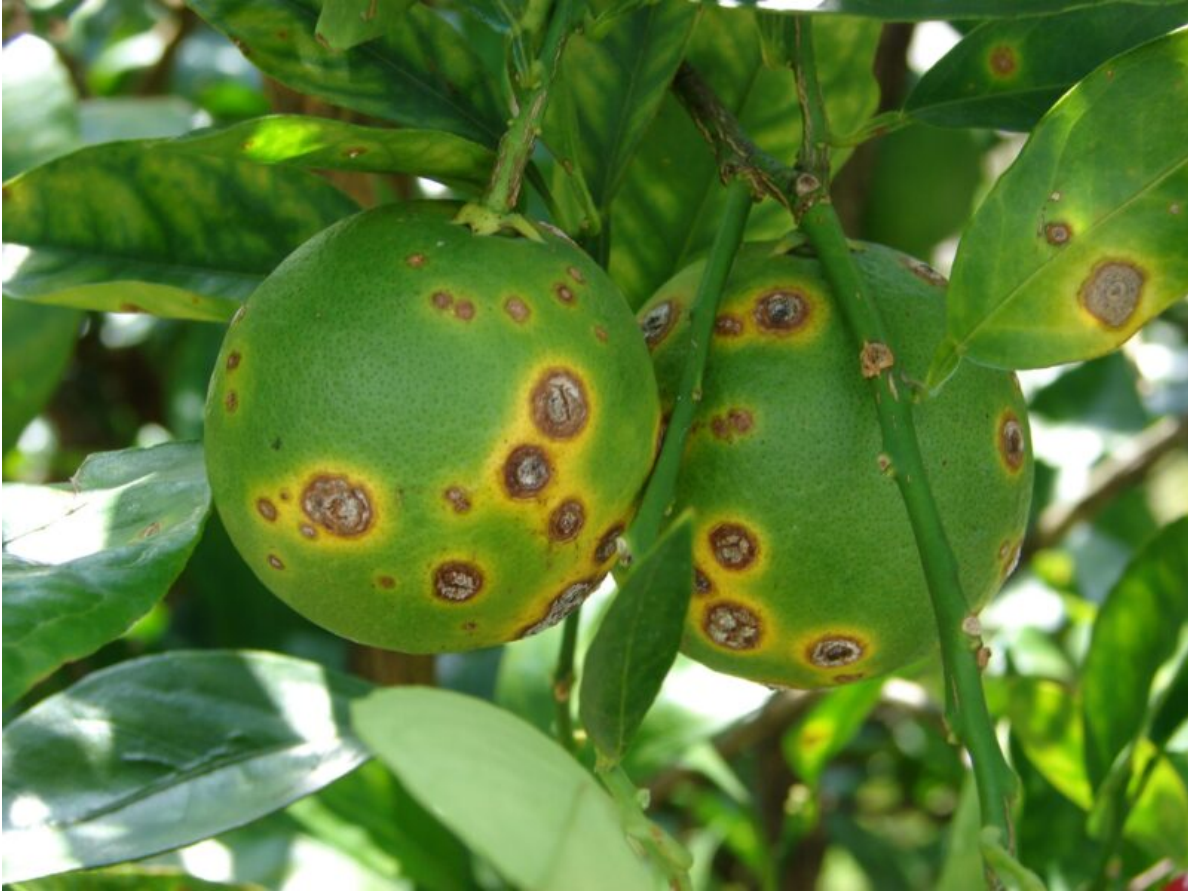


Figure 3: Credits: rawpixel

Computer vision tasks

[Update this]OCR, Pose Detection, Image Generation, Image Captioning, etc. are some of the other tasks in computer vision but we will focus on the following four tasks in this document:

Image classification

- Goal: Assign a label to an image.
- Question: What is the category of the object in the image?
- Input: Images with corresponding labels (e.g., cat, dog, car).
- Example Dataset: [Butterfly Image Classification](#)
- Output: A predicted label for the image.
- Models: Convolutional Neural Networks (CNNs), ResNet, VGG, Inception.

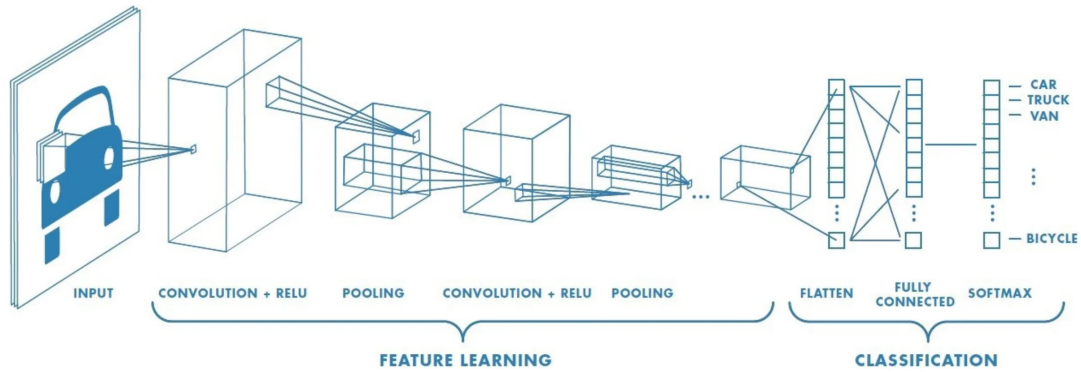


Figure 4: Credits: Necati Demir, Medium

Object detection

- Goal: Identify and locate objects within an image.
- Question: Where are the objects in the image and what are their classes?
- Input: Images with bounding box annotations (x, y, width, height).
- Example Dataset: [Traffic Detection Project](#)
- Output: Bounding boxes and class labels for each detected object.
- Models: YOLO, R-CNN, SSD.

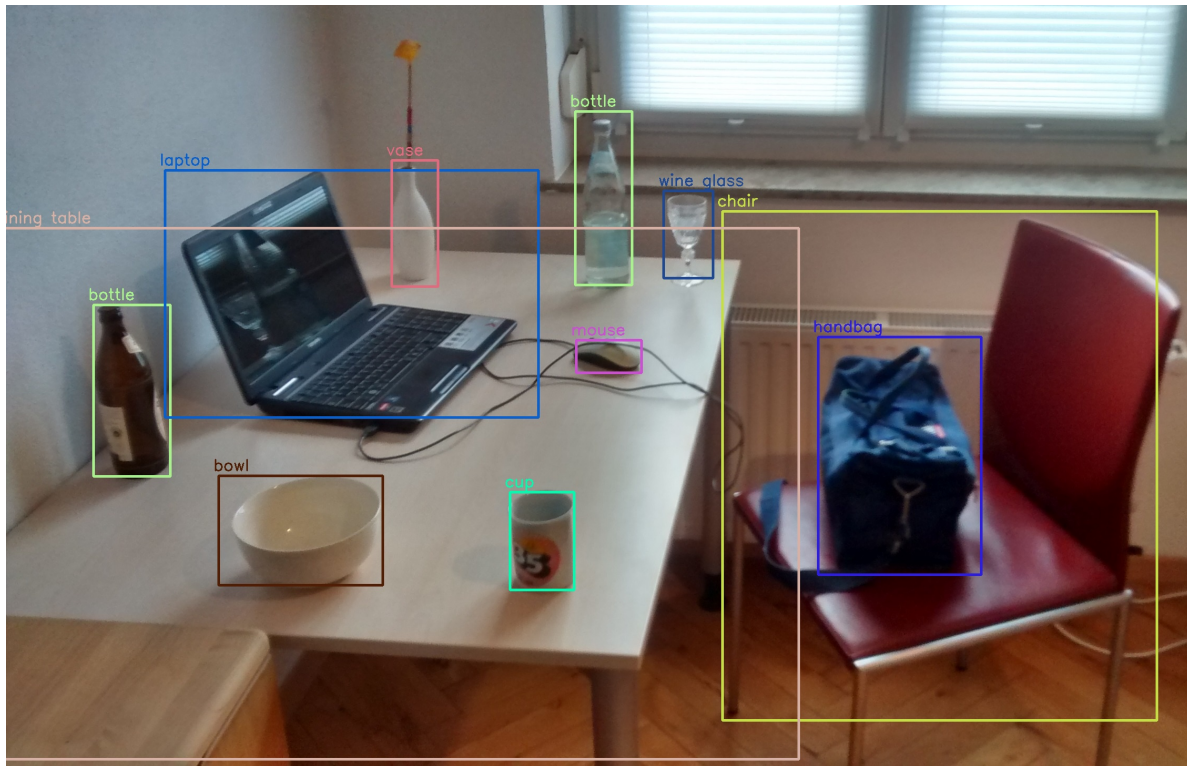


Figure 5: Credits: MTheiler, wikimedia commons

Semantic segmentation

- Goal: Assign a class label to each pixel in the image.
- Question: What is the class of each pixel in the image?
- Input: Images with pixel-level annotations (e.g., road, building, sky).
- Example Dataset: [Cityscapes Image Pairs](#)
- Output: A segmented image where each pixel is labeled with a class.
- Models: U-Net, DeepLab, FCN, SegNet, PSPNet, Mask R-CNN.

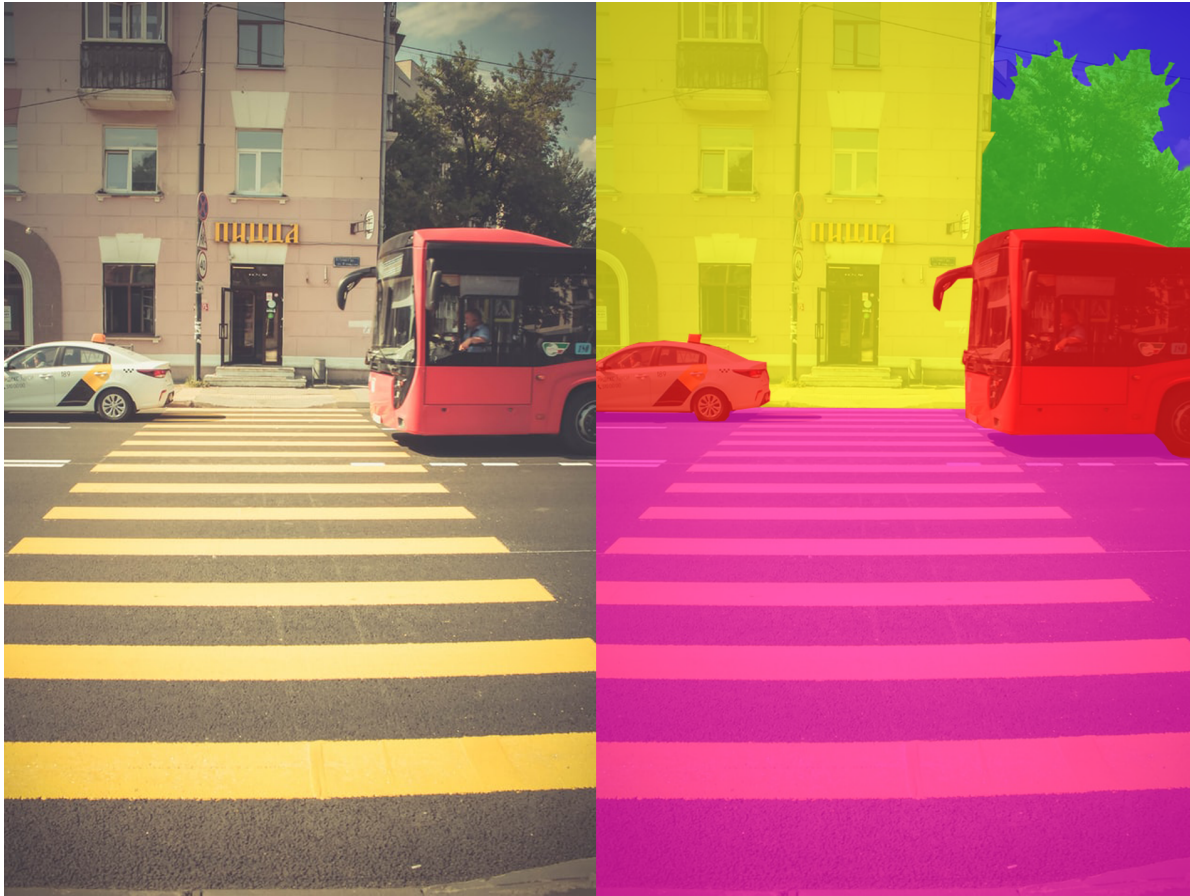


Figure 6: Credits: B.Palac, wikimedia commons

Instance segmentation

- Goal: Identify and segment individual objects in an image.
- Question: What are the boundaries of each individual object in the image?
- Input: Images with instance-level annotations (e.g., each object is labeled separately).
- Example Dataset: [Use API to get the COCO 2018 dataset](#)
- Output: Segmented masks for each individual object in the image.
- Models: Mask R-CNN, YOLOv5-Instance Segmentation.



Figure 7: Credits: <https://kharshit.github.io/>

Comparison of tasks

The key difference in these tasks is the levels of granularity in understanding the image.

- Image classification predicts a single label for an entire image.
- Object detection identifies and localizes multiple objects using bounding boxes while assigning a class label to each detected object.
- Semantic segmentation classifies every pixel in the image according to its category, without distinguishing between separate objects of the same type.
- Instance segmentation goes one step further by classifying each pixel and also separating different instances within the same category.

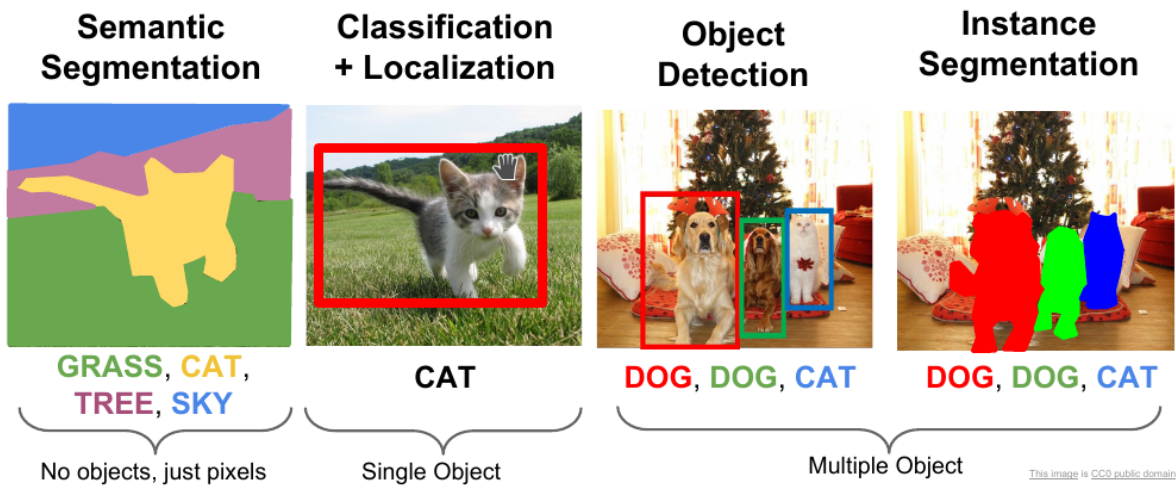


Figure 8: Credits: CC0 public domain

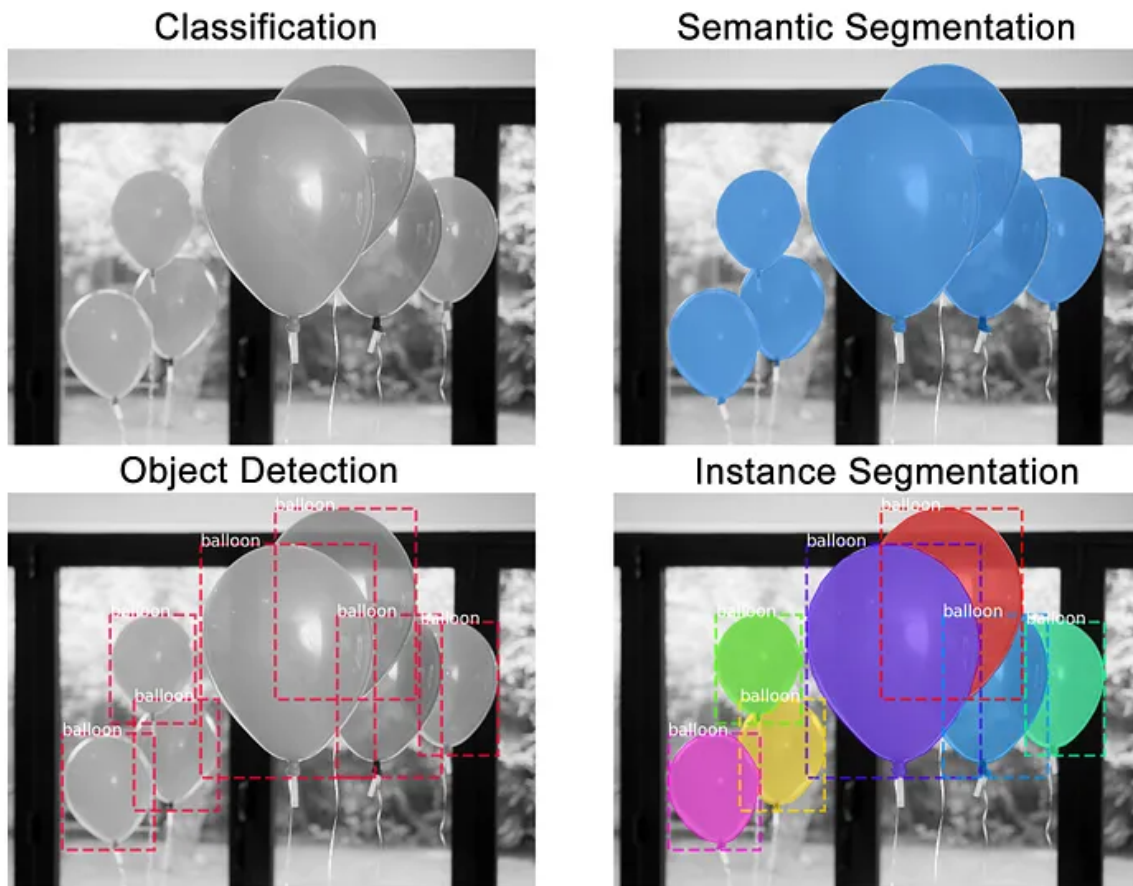


Figure 9: Credits: Vandyahmad24, wikimedia commons

Data

In computer vision tasks, the data typically consists of images (or videos) and their corresponding labels or annotations. The quality and quantity of data play a crucial role in the performance of computer vision models.